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Solution:

According to Boyle's law, at constant temperature for a given mass of gas at constant temperature,

$P_1V_1 = P_2V_2 = P_3V_3$

$1\ atm \times 1\ dm^3 = 2\ atm \times V_2 = P_3 \times 0.25\ dm^3$

$\therefore 2\ atm \times V_2 = 1\ atm \times 1\ dm^3$

$$V_2 = \frac{1\ \cancel{atm} \times 1\ dm^3}{2\ \cancel{atm}}$$

$$\boxed{V_2 = 0.5\ dm^3}$$

and $P_3 \times 0.25\ dm^3 = 1\ atm \times 1\ dm^3$

$$P_3 = \frac{1\ atm \times 1\ \cancel{dm^3}}{0.25\ dm^3}$$

$$\boxed{P_3 = 4\ atm}$$

Evaluate Yourself

1. Freon-12, the compound widely used in the refrigerator system as coolant causes depletion of ozone layer. Now it has been replaced by eco-friendly compounds. Consider 1.5 dm^3 sample of gaseous Freon at a pressure of 0.3 *atm*. If the pressure is changed to 1.2 *atm*. at a constant temperature, what will be the volume of the gas increased or decreased?

2. Inside a certain automobile engine, the volume of air in a cylinder is 0.375 dm^3, when the pressure is 1.05 *atm*. When the gas is compressed to a volume of 0.125 dm^3 at the same temperature, what is the pressure of the compressed air?

6.2.2 Charles Law (Volume-temperature relationship)

The relationship between volume of a gas and its temperature was examined by J. A. C. Charles. He observed that for a fixed mass of a gas at constant pressure, the volume is directly proportional to its temperature (K). Mathematically it can be represented as (at constant P and n)

$$\text{or} \qquad V = kT \quad \text{-------} \quad (6.5)$$

$$\text{or} \qquad \frac{V}{T} = \text{Constant}$$

If the temperature of the gas increases, the volume also increases in direct proportion, so that $\frac{V}{T}$ is a constant. For the same system at constant pressure, one can write

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \text{Constant} \quad \text{-----} \quad (6.6)$$

For example, if a balloon is moved from an ice cold water bath to a boiling water bath, the temperature of the gas increases. As a result, the gas molecules inside the balloon move faster and gas expands. Hence, the volume increases.



Ice water Boiling water

Figure. 6.4 Air filled balloon in ice cold and hot water

Variation of volume with temperature at constant pressure

The plot of the volume of the gas against its temperature at a given pressure